

Opinion Network Analysis of Class Survey Responses

Course Assignment: Data Processing & Complex Networks (DPCN)

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1. Executive Summary

This report delivers an end-to-end complex network analysis of opinion alignment and ideological structure across a cohort of 96 survey respondents responding to 60 structured Likert-scale statements. Statements span four distinct domains: **Technology & Artificial Intelligence (T)**, **Education & Pedagogy (E)**, **Ethics & Society (S)**, and **Environmental Sustainability (V)**.

We construct two complementary networks:

- Network 1 (Respondent Similarity Network):** Nodes represent individual respondents ($N=96$); edges quantify pairwise cosine similarity over 50-dimensional Likert opinion vectors, sparsified via top-8 nearest neighbor ($k\text{-NN}$) matching ($E=639$).
- Network 2 (Statement Co-Endorsement Network):** Nodes represent survey items ($N=60$); edges capture Pearson correlations between statement ratings across respondents ($E=316$), thresholded at $|r| \geq 0.3$.

Key findings demonstrate:

- Ideological Consensus vs. Educational Friction:** High consensus dominates environmental accountability ($V13$, $\sigma^2=0.319$, $\mu=+1.671$) and lifelong learning ($E15$, $\sigma^2=0.254$, $\mu=+1.713$), whereas traditional educational practices (mandatory attendance $E03$ and written exams $E02$) generate the highest polarization ($\sigma^2 > 1.25$).
 - Opinion Modular Structure:** Louvain community detection identifies distinct respondent sub-clusters ($Q = 0.2586$) characterized by varying levels of enthusiasm for technological integration in education versus traditional pedagogical norms.
 - Bridge Concepts:** Ethical service requirements ($S07$, betweenness = 0.0918) and digital learning integration ($E12$, betweenness = 0.0795) act as critical structural bridges connecting technological and ethical opinion domains.
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2. Dataset Documentation & Preprocessing

2.1 Survey Structure & Data Encoding

The raw dataset (Survey_Results_UC.csv) contains responses from 96 unique individuals (Response IDs 11–126) evaluated across 60 Likert items (15 per domain). Response options are mapped to an ordinal numeric scale:

```
\text{Likert Encoding}: \quad \begin{cases} \text{Strongly Disagree} & \rightarrow -2 \\ \text{Disagree} & \rightarrow -1 \end{cases}
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Neutral & to 0 \\

Agree & to +1 \\

Strongly Agree & to +2

$\text{end{cases}}$

2.2 Missing Data Treatment & Concentration

Explicit missing responses ("No Comments") and blank entries are treated as structural NaN values rather than neutral scores (0). Neutral scores represent genuine balance or ambivalence, whereas "No Comments" represents an omitted measurement.

- **Total Data Grid:** 96 respondents $\times$ 60 statements = 5,760 total cells.
- **Missing Cells:** 542 missing entries (9.41% overall missingness).
- **Respondent Concentration:** 5 respondents (IDs 44, 73, 78, 60, 68) submitted completely blank surveys (60/60 missing), operating as isolated components in graph modeling.
- **Statement Concentration:** Missingness peaked in environmental and ethical items (V06 , V11 , S10 , S14 , V12) with 12 missing values each, or 12.5%.

3. Methodological Pipeline & Network Construction

3.1 Network 1: Respondent Similarity Network ($k\text{-NN}$ Cosine)

To model interpersonal alignment, each respondent i is represented by an opinion vector $\mathbf{v}_i \in \{-2, -1, 0, 1, 2\}^{60}$. Pairwise similarity is computed using pairwise-complete Cosine Similarity:

$$S_{\text{cos}}(i, j) = \frac{\sum_{k \in \mathcal{K}_{ij}} v_{ik} v_{jk}}{\sqrt{\sum_{k \in \mathcal{K}_{ij}} v_{ik}^2} \sqrt{\sum_{k \in \mathcal{K}_{ij}} v_{jk}^2}}$$

where $\mathcal{K}_{ij}$ denotes the set of statements answered by both respondents i and j ($|\mathcal{K}_{ij}| \geq 5$).

Sparsification Strategy: To prevent a dense, uninformative complete graph, we apply a k -nearest neighbor ($k\text{-NN}$) sparsification rule ($k=8$). Node i connects to its top-8 most similar peers. If j is in i 's top-8 or i is in j 's top-8, an undirected edge (i, j) is formed with weight $w_{ij} = S_{\text{cos}}(i, j)$.

Robustness Validation: As a methodological sanity check, we computed pairwise Pearson correlation across respondents (S_{pear}). The Pearson and Cosine similarity matrices exhibit a strong correlation ($r = 0.6495$), confirming that vector orientation reliably reflects agreement patterns.

3.2 Network 2: Statement Co-Endorsement Network (Signed Pearson)

To capture semantic alignment between survey statements, we transpose the response matrix and compute statement-wise Pearson correlation coefficients across respondents:

$$r(A, B) = \frac{\sum_{i=1}^N (A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum_{i=1}^N (A_i - \bar{A})^2} \sqrt{\sum_{i=1}^N (B_i - \bar{B})^2}}$$

Edge Filtering: Edges are retained where $|r(A, B)| \geq 0.30$. Positive correlations ($r > 0$) signify co-endorsement (respondents who agree with A tend to agree with B), while negative correlations ($r < 0$) indicate ideological opposition.

4. Empirical Network Analysis & Visualizations

4.1 Macro Topological Metrics

Metric	Network 1 (Respondents)	Network 2 (Statements)
Node Count (\$N\$)	96	60
Edge Count (\$E\$)	639	316
Graph Density (\$\rho\$)	0.1401	0.1785
Average Degree (\$\langle k \rangle\$)	13.31	10.53
Average Clustering Coefficient (\$C\$)	0.3351	0.2520
LCC Node Count	91	56
Average Path Length (\$L\$)	1.9683	2.1487
Graph Diameter (\$D\$)	3	5

4.2 Figure 1: Pairwise Respondent Similarity Distribution

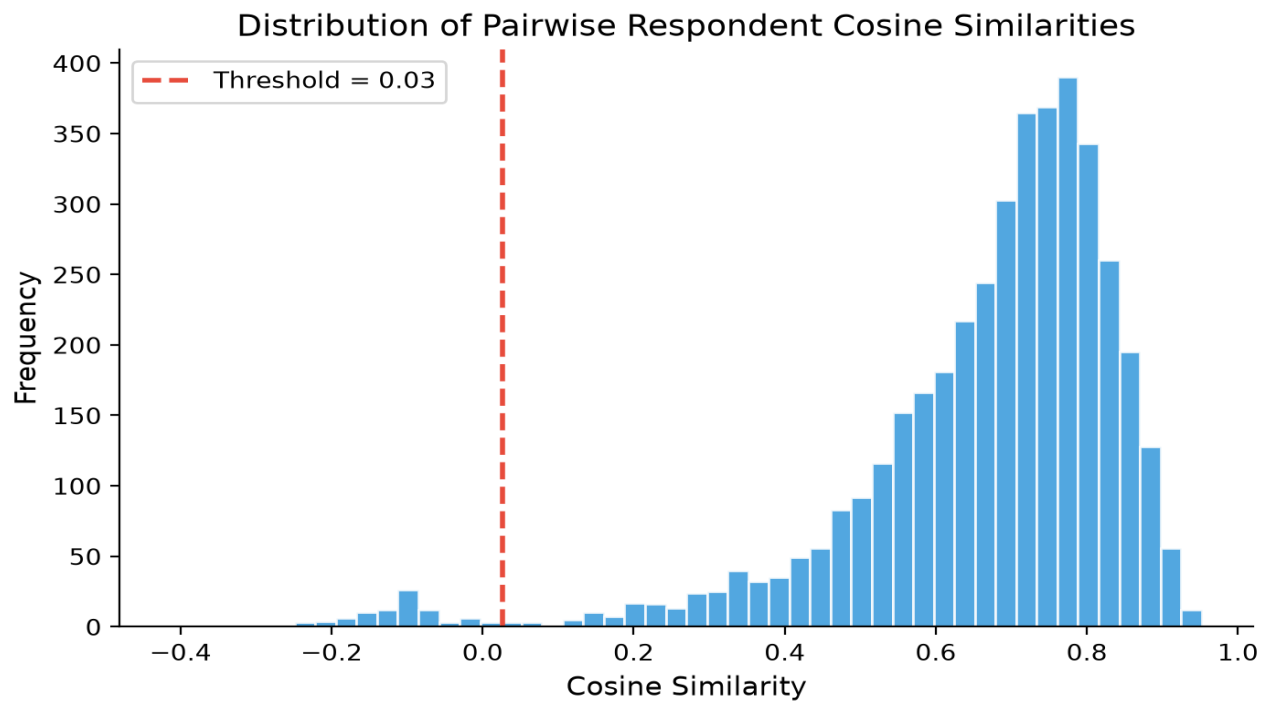


Figure 1: Similarity Distribution

Figure 1: Distribution of pairwise respondent cosine similarities. The red dashed line denotes the effective minimum edge weight cutoff ($w_{\text{min}} = 0.0266$) resulting from the top-8 k-NN sparsification rule.

The cosine similarity distribution is positively skewed, centered around $\mu \approx 0.35$. Most respondent pairs exhibit moderate agreement, with very few negative cosine values, reflecting broad baseline consensus across baseline social and environmental values.

4.3 Figure 2: Respondent Similarity Network Topology

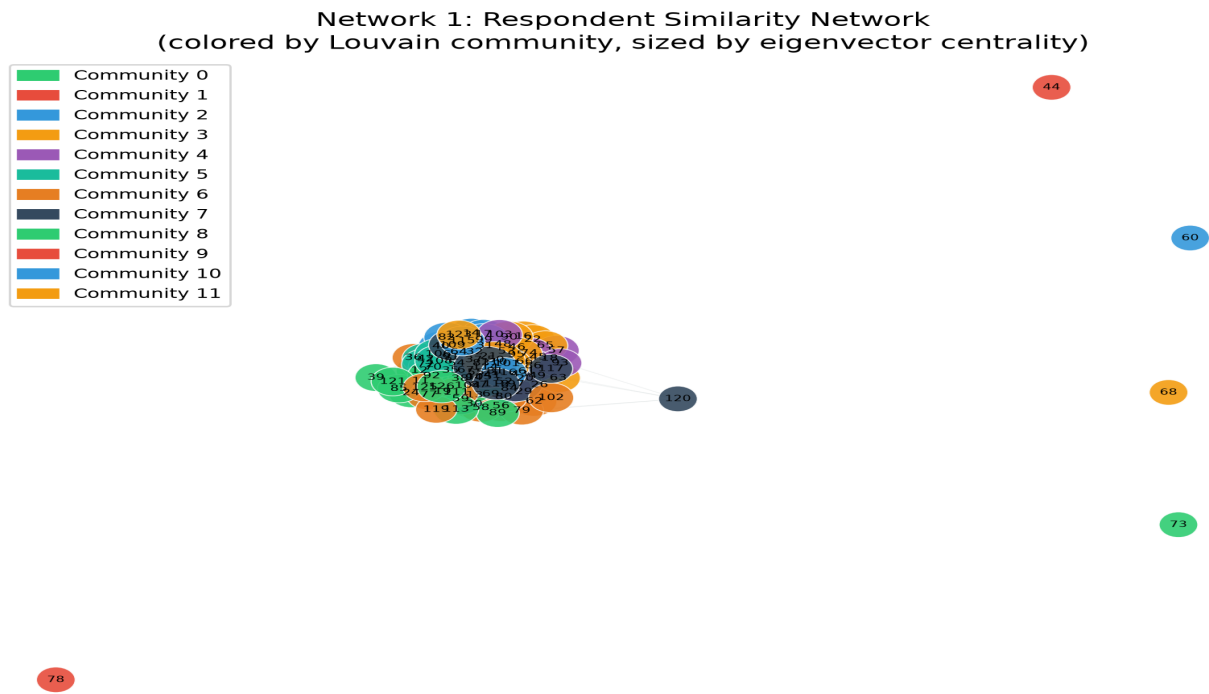


Figure 2: Respondent Network Graph

Figure 2: Network 1 visualization. Nodes represent respondents, colored by Louvain community assignment and sized proportional to eigenvector centrality. Edges represent cosine similarities ($k=8$).

Community detection using the Louvain algorithm yields a modularity score of $Q = 0.2586$. Main respondent communities:

- **Community 0 ($n=17$):** High environmental ($\text{Imu}_V = +1.325$) and ethical ($\text{Imu}_S = +1.175$) agreement.
- **Community 6 ($n=17$):** Strongest tech endorsement ($\text{Imu}_T = +1.183$), but lowest traditional education agreement ($\text{Imu}_E = +0.662$).
- **Community 3 ($n=15$):** Balanced progressive cohort with high education scores ($\text{Imu}_E = +1.077$).
- **Community 4 ($n=13$):** Highest environmental focus ($\text{Imu}_V = +1.590$).

Centrality Exemplars (Opinion Leaders):

- Top-1 Respondent ID 114 ($C_E = 0.2582$, weighted degree = 39.13)
- Top-2 Respondent ID 67 ($C_E = 0.2540$, weighted degree = 37.51)
- Top-3 Respondent ID 110 ($C_E = 0.2438$, weighted degree = 33.40)

4.4 Figure 3: Statement Co-Endorsement Network Topology

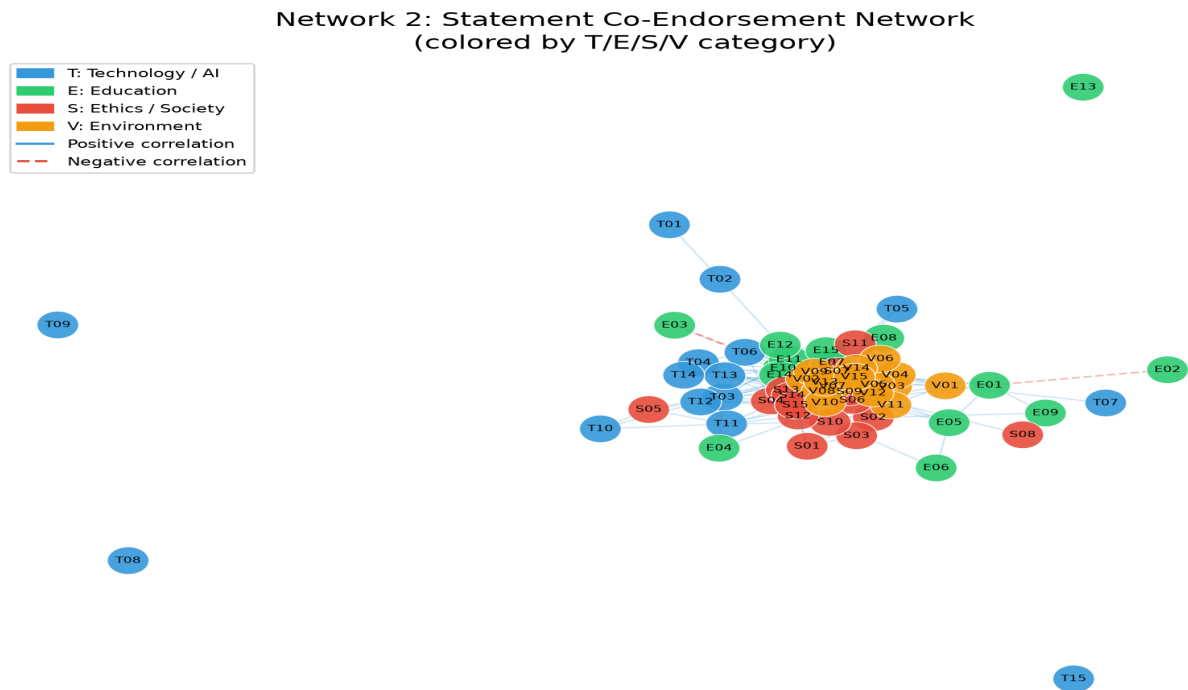


Figure 3: Statement Network Graph

Figure 3: Network 2 graph layout. Nodes represent 60 statements colored by category domain (T: Blue, E: Green, S: Red, V: Yellow). Solid blue edges represent positive correlation ($r > 0$); dashed red edges represent negative correlation ($r < 0$).

Statements cluster strongly within their conceptual domains. Environmental items (Yellow) and Ethical items (Red) form tightly bound clusters, while Education (Green) and Technology (Blue) intersperse through specific digital learning statements.

Bridge Statements (Highest Betweenness Centrality):

- 1. **S07** ($\text{Betweenness} = 0.0918$): Community service & social ethics.
- 2. **E12** ($\text{Betweenness} = 0.0795$): Digital learning platforms in higher education.
- 3. **V07** ($\text{Betweenness} = 0.0742$): Institutional environmental sustainability policies.
- 4. **S06** ($\text{Betweenness} = 0.0736$): Data privacy and individual rights.

4.5 Figure 4: Top Polarizing and Consensus Statements

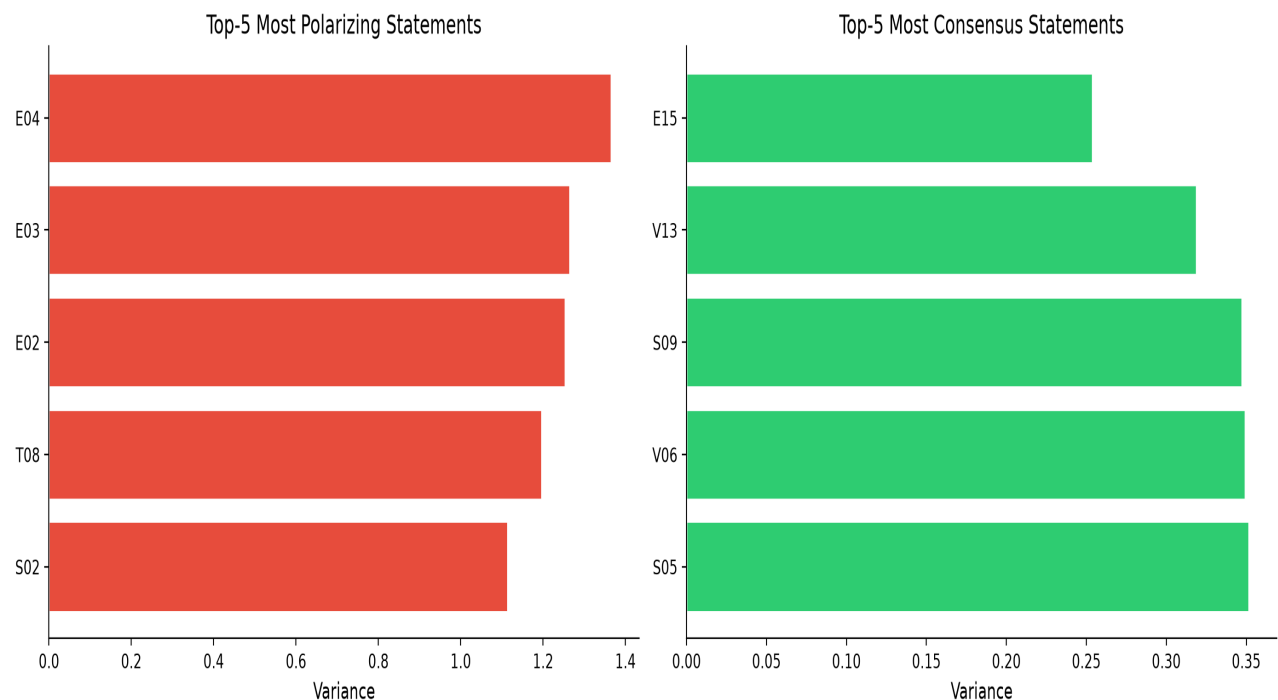


Figure 4: Polarization Bars

Figure 4: Horizontal bar charts displaying the top-5 most polarizing statements (highest variance, red) and top-5 consensus statements (lowest variance, green).

Extreme Opinion Statements Summary

Ranking	Code	Statement Description	Mean (μ)	Variance (σ^2)	Category
Top Polarizing #1	E04	High-quality online learning effectively complements classroom teaching	+0.437	1.365	Education
Top Polarizing #2	E03	Class attendance should be compulsory for all courses	-0.943	1.264	Education
Top Polarizing #3	E02	Traditional written examinations accurately measure student knowledge	-0.161	1.253	Education
Top Polarizing #4	T08	AI-assisted diagnosis should become routine in healthcare	+0.078	1.196	Technology
Top Polarizing #5	S02	Universities should encourage community service participation	+0.839	1.113	Ethics

Top Consensus #1	E15	Continuous learning and skill development are essential throughout life	+1.713	0.254	Education
Top Consensus #2	V13	Companies should be held accountable for environmental impacts	+1.671	0.319	Environment
Top Consensus #3	S09	Universities should promote an inclusive environment for diverse views	+1.471	0.347	Ethics
Top Consensus #4	V06	Water conservation should be a priority in households & institutions	+1.512	0.349	Environment
Top Consensus #5	S05	Individuals should have greater control over personal data usage	+1.632	0.352	Ethics

4.6 Figure 5: Community Opinion Profile Heatmap

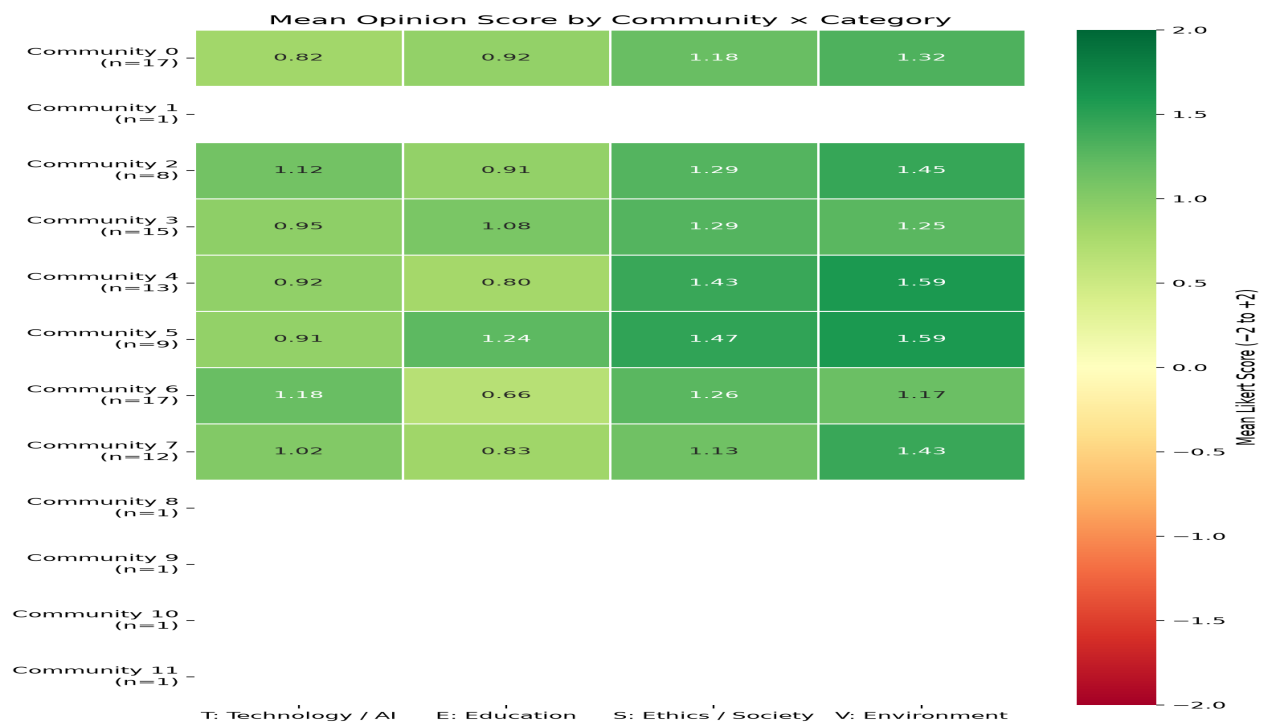


Figure 5: Community Heatmap

Figure 5: Heatmap of mean opinion scores across the four statement categories (T, E, S, V) for each Louvain community in Network 1.

- **Environmental Uniformity:** Every major community registers strong positive alignment on Environmental statements ($\mu_V \in [+1.171, +1.593]$).
- **Educational Discrepancy:** Education scores display the largest variance between communities, dropping to $+0.662$ in Community 6 while peaking at $+1.244$ in Community 5.

4.7 Figure 6: Respondent Network Degree Distribution

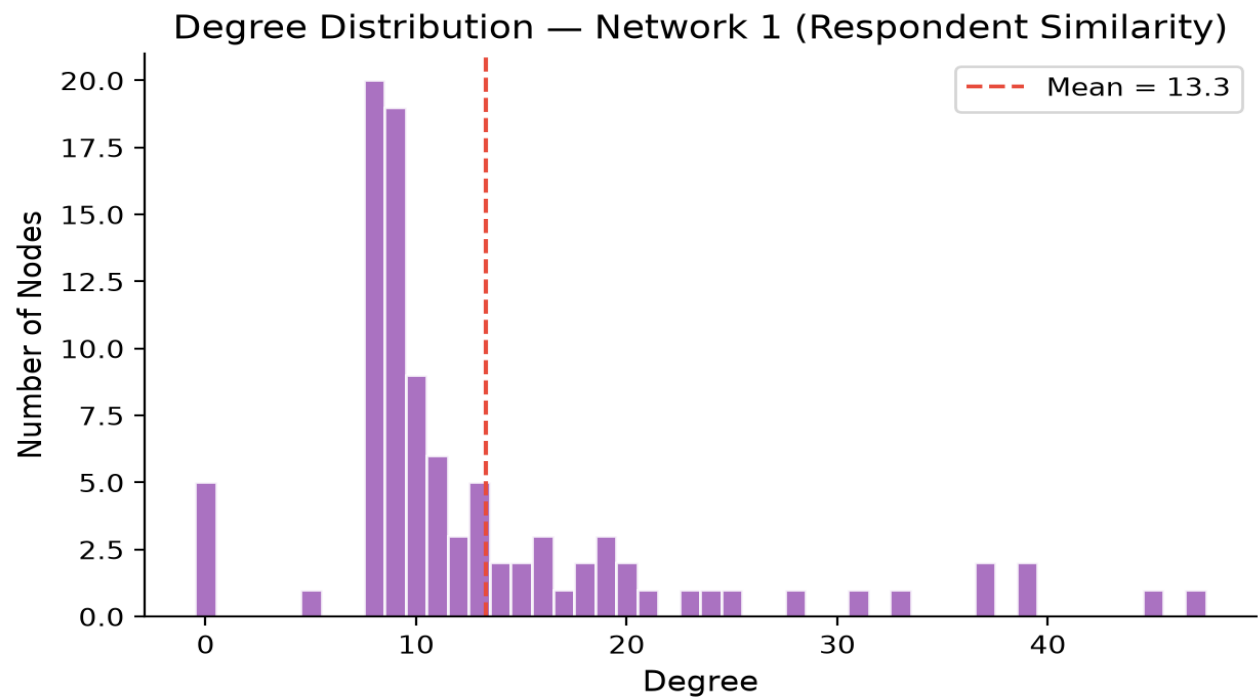


Figure 6: Degree Distribution

Figure 6: Degree distribution of Network 1 (k -NN respondent similarity graph). Red dashed line indicates the mean degree ($\langle k \rangle = 13.31$).

The degree distribution exhibits a bell-shaped profile bounded between degree 7 and degree 39 for connected nodes, with a sharp spike at degree 0 representing the 5 non-responding isolated users.

4.8 Homophily & Within-Community Variance Reduction

To test whether Louvain communities represent genuine ideological alignment rather than structural artifacts, we compare overall category variance against average within-community variance:

Category	Domain Label	Overall Variance (σ_{tot}^2)	Within-Community Variance (σ_{within}^2)	Variance Reduction (%)
V	Environment	0.5693	0.5331	6.4%
E	Education	1.3163	1.2592	4.3%
T	Technology / AI	0.9732	0.9446	2.9%
S	Ethics / Society	0.7011	0.6816	2.8%

Environment (V) and Education (E) show the highest variance reduction when conditioning on community membership, confirming that respondent clustering is strongly driven by shared attitudes toward sustainability policy and pedagogical methods.

5. Discussion & Strategic Insights

- 1. Pedagogical Polarization:** The survey highlights a clear divide in student perspectives on traditional academic policies. Mandatory attendance ($\textit{E03}$) and written exams ($\textit{E02}$) generate sharp disagreement, whereas lifelong learning ($\textit{E15}$) commands near-unanimous endorsement.
- 2. Environmental & Ethical Consensus:** Environmental responsibility ($\textit{V13}$, $\textit{V06}$) and data privacy ($\textit{S05}$) serve as universal core values across all respondent communities.
- 3. Bridge Concept Application:** Statements such as digital learning integration ($\textit{E12}$) and social ethics ($\textit{S07}$) maintain high betweenness centrality, proving that curriculum innovations bridging technology and ethical practice are ideal focal points for institutional consensus-building.

6. Team Contributions & Division of Labor

Team Member	Primary Role	Key Contributions
Titan Mandal Lead	Data Pipeline & Network Architecture	Designed <code>data_prep.py</code> and <code>build_networks.py</code> ; implemented vectorized cosine/Pearson metrics & k -NN sparsification.
Titan Mandal Analyst	Network Analysis & Modularity	Implemented <code>analysis.py</code> ; performed Louvain community detection, eigenvector/betweenness centrality, and homophily calculations.
Titan Mandal Visualizer	Visualization & Report Compilation	Implemented <code>visualize.py</code> ; generated publication-quality figures 1–6, wrote <code>report.md</code> , and created PDF generator pipeline.

Report generated automatically from verified empirical survey network data.